

QTLs underlying inflorescence architecture in sorghum (*Sorghum bicolor* (L.) Moench) as detected by association analysis

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Abstract The pattern of the panicle is an important characteristic of sorghum that is used to identify race divisions and species values. The knowledge of the genetic basis of sorghum inflorescence architecture and its component traits can enhance the process of improvement in sorghum breeding. This study was undertaken to better understand the genetic basis of sorghum inflorescence architecture and its association with yield-related traits. We investigated 98 simple sequence repeat maps in the sorghum diversity research set (SDRS) of 107 landraces from the worldwide sorghum germplasm. A significant difference between accessions was observed for 14 measured traits. In several components of sorghum inflorescence architecture, we found that the

significance and high correlations between pairs of traits revealed that these variations are dependent not only on the panicle length (PanL) but also on the total branch number, maximum length of the primary branch (MaxLBZ/MxLBZ), rachis length (Rac), and panicle diameter (PanD) and width (Panw). Significant mean squares were obtained for almost all characters in the individual analysis and combined analysis of variance for each season and across two seasons, suggesting that, the components of inflorescence architecture and yield related traits were highly variable in SDRS population, therefore, would respond to selection for yield improvement. Molecular markers divided the germplasm into three sub-populations with different groups of inflorescence/panicle types (PanT) (broom, open and compact). The SDRS was composed of six different PanT that were associated with 14 inflorescence traits. Using different models of association analysis, 16 loci on 8 chromosomes were found to be significantly related to the patterns of observed panicle traits. Among these loci, four QTLs were responsible for length-based traits, and three QTLs were responsible for dimensional traits. Four QTLs were responsible for panicle feature such as PanT and shape, one QTL responsible for yield related trait such as grain weight. Two QTLs were responsible for branch-based traits, such as the total node number and branch number with $-\text{Log}_{10}(P)$ values ranging from 1.5 to 7.6 as the threshold value. Several genomic regions affected multiple traits, including one region that affected PanL and MxLBZ/MaxLBZ. QTLs for

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different traits tended to be found in the same region on chromosome 4 and on chromosome 9. Additionally, QTL was involved on Chr-2, Chr-5, Chr-6 and Chr-10 as a novel QTL that controlled Rac, total node number, PanD and PanT. These results will serve as preliminary findings of QTLs for sorghum inflorescence architecture, and a further evaluation of the germplasm for these traits is in progress. Moreover, these findings can lead to the emergence of a new era of sorghum genomics, which are aimed at bridging the knowledge gap between genotype and phenotype in sorghum inflorescence architecture.

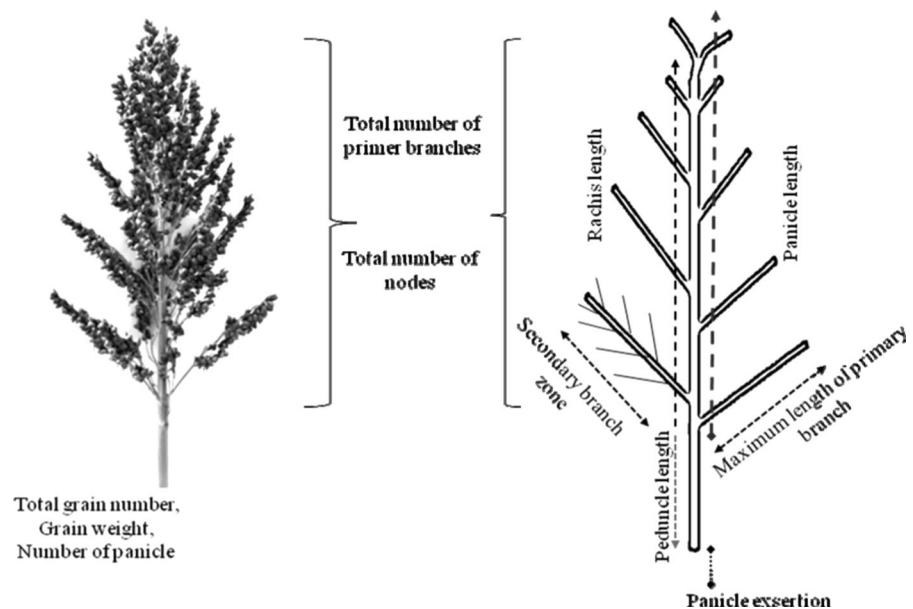
Keywords Genome-wide association mapping · Inflorescence architecture · Panicle traits · *Sorghum bicolor* · SSRs · Yield-related traits

Introduction

Throughout the world, grass species are economically important as both staple grain foods for humans and feedstock for animals. Sorghum (*Sorghum bicolor* (L.) Moench) is the fifth most important C₄ cereal crop globally (FAO 2012; Delserone 2007) that can survive the harsh climatic conditions of arid and tropical environments. Unlike rice and other staple crops, which are widely used for food and industrial purposes, sorghum has thus far remained a traditional food crop of subsistence farmers (Rai et al. 1999; Singh et al. 2001). Cultivated sorghum is classified into five main races (bicolor, guinea, caudatum, durra and kafir) (Barnaud et al. 2008; Harlan and de Wet 1972). Sorghum types can be identified according to their morphological traits (Kaitaniemi et al. 1999). The identification of racial divisions and species values are primarily based on panicle and grain characters (Abdi et al. 2002; Harlan and de Wet 1972; Murray et al. 2009). Sorghum panicles, which are called inflorescences, show remarkable diversity in morphological, physiological, genetic and ecological traits (House 1985; Barkworth 2003). Panicles have a diverse set of morphologies and complex morphological characteristics. The basic structure of the mature panicle is a head that is supported by a stem (peduncle), which is usually straight but is curved in some cases (Fig. 1). The main axis is the rachis, which runs the length of the panicle from the bottom branch

(whorl) to the top. Several branches (whorls) arise from the internodes of the rachis. Each lateral may also branch repeatedly, with each primary branch dividing into secondary branches, and tertiary branches originating from secondary branches. The final branches then carry the spikelets, which can be single or consist of several paired spikelets (Harlan and de Wet 1972; House 1985; Dahlberg 2000; Barkworth 2003). Sorghum inflorescence architecture is not only an important factor for sorghum identification, which contributes to both the yield and quality of sorghum, but is also an important determinant of rice and maize tassels because of its close associations with grain yield and grain quality, which include several commercially important traits (Bommert et al. 2005; Ikeda et al. 2010; Yan et al. 2007). Breeders have greatly improved plant architecture, potential energy and grain yield productivity. Panicle morphology can directly affect grain yield; therefore, knowledge of the genetic basis of sorghum inflorescence architecture and its components can complement the breeder's efforts to improve sorghum. Of these characteristics, sorghum inflorescences have become a new model system for functional genomics from agronomic, developmental, and evolutionary viewpoints similar to other important model cereal crops (Doggett 1988; Emma et al. 2013), such as maize and rice; however, a few morphological characteristics of inflorescence architecture have been mapped similar to major effect genes across a range of genetic linkage maps (Colasanti et al. 1998; Harlan and de Wet 1972; Ikeda et al. 2010). Genome-wide association mapping (GWAS) has been widely applied in identifying the causal variants that affect complex agronomical traits in other cereal crops. The use of GWAS in sorghum remains new and has a great potential for a variety of functions in crop improvement and linkage map construction. The inflorescence pattern of sorghum is a complex phenotype, which involves many genes. In sorghum, most yield and yield-related traits are polygenic; however, inflorescence architecture remains the most polygenic and complex trait (House 1985; Bello et al. 2001; Zou et al. 2011). The genetic control of polygenic and complex traits is primarily quantitative. Moreover, GWAS have formed the genetic basis of agronomic traits in sorghum (Bouchet et al. 2012; Brown et al. 2006; Casa et al. 2005; Shehzad et al. 2009b); however, compared with maize and rice, the genome-wide association of sorghum inflorescence

Fig. 1 Scheme of sorghum inflorescence (panicle) traits that were investigated



architecture has not been studied extensively thus far. The critical point of the genetic basis of sorghum inflorescence architecture has been one of the major scientific challenges in the process of sorghum crop improvement. Therefore, we aimed to measure the intraspecific variation in panicle traits across the 107 landraces in the sorghum diversity research set (SDRS) (Shehzad et al. 2009a) to characterize not only a wide range of genetic diversity, as well as phenotypic diversity, and its suitability for association analyses. Fourteen sorghum panicle and panicle component traits were used to confirm the high level of selectivity of the approach. The values of these traits, correlations to the yield components and preliminary efforts for the association mapping analysis may be useful information for modern sorghum plant breeding. Therefore, the objectives of this study were (1) to gain an understanding of the variation in sorghum inflorescence architecture and (2) to investigate the genetic basis of inflorescence architecture and its association with yield-related traits.

Materials and methods

Experimental design and field procedure

The study was conducted during the FY 2010 and FY 2011 sorghum main growing season in the

experimental field at the Agricultural and Forestry Centre, University of Tsukuba, Japan. All accessions were planted under field condition is located in Ibaraki ken, Japan (36.07°/101.71° latitude and 140.05°/40.24° longitude). Dried seeds were sterilized in fungicide 1 week before sowing and seeds were sown manually by dibbling method directly into the field plots. Seeds are dibbled into a 2–3 cm deep and 1 cm apart in one hole. Four individual plants were grown out for each of the 107 accessions, and plants from the different accessions were grown together in the same field in 6 m × 1 m plots. Single row plots measuring 6 m long with between row spacing of 1 m were used for each replication. Weed control was done around plots in every week during crop season. Regular irrigation, fertilizer and nutrients were used at 35 days after germination. Disease and insects were controlled with fungicide and insecticide. At the anthesis stage the duration in days from planting when 50 % of the plants within a plot were shedding pollen (Vanderlip and Reeves 1972; IBPGR and ICRISAT 1993), all panicles were covered by paper bags to prevent outcrossing and several damages. The bags were removed at the maturity grain stage. From all labeled plants mature panicles were harvested and dried naturally with net bags before traits measurement. Scoring of phenotypic characters was evaluated on two replications in each year (FY 2010 and FY 2011).

Plant materials and trait measurements

In this study, we used a previously selected SSR based SDRS of 107 landraces from 27 countries in Asia and Africa. The SDRS was developed from a geographically diverse based population of 320 sorghum landraces by the assessment of 38 SSR markers which were randomly selected from ten chromosomes of sorghum (Shehzad et al. 2009a). There accessions were selected from more than 3,500 germplasm conserved in Genebank, National Institute of Agrobiological Sciences, Japan. We categorised the plant material into three different groups, such as the East Asian group (25 accessions), the other regions of Asia group (two from Southeast Asian, 26 from South Asian, two from Southwest Asian) and the African group (52 accessions from African) according to previous research by Shehzad et al. (2009b) [Electronic Supplementary Material (ESM) Table 1]. Data were recorded on four randomly selected plants samples from each replication. Two categories data were recorded for 14 panicle traits (11 quantitative traits based on metric scale and 3 qualitative traits based on an arbitrary scale) according to the sorghum descriptors from IBPGR, ICRISAT (1993) and NIAS, as well as Genebank descriptors (see details in ESM Table 2).

The basic structure of the mature panicle is a head that is supported by a stem (peduncle), which is usually straight but is curved in some cases. The main axis is the rachis, which runs the length of the panicle from the bottom branch to the top. Several branches (whorls) arise from the internodes of the rachis. Each lateral may also branch repeatedly, with each primary branch dividing into secondary branches, and tertiary branches originating from secondary branches. The final branches then carry the spikelets. At the mature stage, main component panicle traits, including the rachis length (Rac), panicle length (PanL), peduncle length (Pend), panicle shape (PanS), and panicle type (PanT) were evaluated (Fig. 1). The panicle diameter (PanD) was measured with a digital vernier calliper (Digimatic Solar DC-S15 m, Mitutoyo, Japan). The Rac was measured as the distance from the bottom whorl to the topmost whorl. The Pend was measured as the distance from the flag leaf to the lowest primary branch zone. After harvesting, all labelled panicles were dried and cleaned before trait measurement. After cleaning, the total number of nodes (TotN), total

number of primary branches (TotBr) and the maximum length of primary branches (MaxLBZ/MxLBZ) were manually measured at the basal part of the panicle. TotBr were removed and counted individually. The TotN (total number of the whole on the rachis) was counted along main axis. For the maximum length of the branch zone (MaxLBZ), three random branches were selected from the longest branch zone in the bottom third portion of the panicle. Next, we evaluated the actual number of grains per panicle (GNP), with an average of three panicles in the accession, and the grain weight per panicle (GWP) from each panicle. All grains were threshed and measured for GWP and GNP traits. Additional data regarding ordinal grouping observations and characteristics, i.e., plant height (PanH), panicle broadness (PanB), neck length (NecL), panicle width (PanW), awn presence (Awn) and awn length (AwnL) were also recorded; (data were not shown). The details of the traits that were measured in this study, as well as the stages of measurement, are explained and listed in ESM Table 2. All primary panicle traits and panicle-related traits were evaluated in three panicles per each accession of each replication. In total, six panicles per each accession of two replications were measured. Yield traits were measured in three labelled plants of each accession per row, which included the total panicle number per plant.

PCR amplification and gel electrophoresis

The SDRS was genotyped using 98 SSR markers from Shehzad et al. (2009b). The SSRs were selected from published linkage maps of sorghum, which were revealed by Bhattaramakki et al. (2000), Kong et al. (2000), and Taramino et al. (1997). In this study, 98 SSR maps were used from previous linkage maps from Shehzad et al. (2009a). The list of total sorghum microsatellite markers with the chromosome location, sequence information, size range and other information are provided in ESM Table 3. DNA samples were extracted from the leaves of 40 days old plants using the CTAB method as described by Murry and Thompson (1980), with some modification. Ninety-eight microsatellite markers were chosen for the analysis. The extraction buffer was composed of 2 % CTAB, 50 mM Tris-HCl (pH 8.0), 10 mM EDTA, 0.7 M NaCl, 0.1 % Proteinase K, 2 % insoluble PVP and 2 % 2-mercaptoethanol. A chloroform-isoamyl

alcohol (24:1 v/v) extraction was performed to remove the cellular debris and proteins. The DNA was precipitated by adding 2-propanol, and the precipitate was rinsed with 70 % and then 95.5 % ethanol. The final precipitate was dissolved in 50 µl 1/10 TE solution and stored at 4 °C. The DNA concentration was measured using a NanoDrop ND-1000 (Thermo scientific) spectrometer and diluted to a working concentration of 5 ng/µl.

PCR amplification of the sorghum SSRs were performed in a 10 µl reaction mixture, which contained 10 ng DNA template, 10× PCR buffer (Mg²⁺ concentration: 20 nM), 2 mM dNTPs, 25 ng of each primer and 0.02 U of the Taq polymerase (Toyobo Co., LTD., Japan) enzyme, using Applied Biosystems 9700 and 2700 thermal cyclers. The annealing temperature was determined for all primers using an Eppendorf Master Cycler EP Gradient S. The PCR was conducted using the following setup profile: denaturation at 94 °C for 5 min, followed by 35 cycles at 94 °C for 1 min, SSR 55/SSR 61/SSR 67 °C for 1 min, 72 °C for 2 min, with a final extension at 72 °C for 5 min and then cooling at 4 °C. PCR products were fractionated through a 10 % (w/v) polyacrylamide gel (10 cm in size) with a constant supply of 200 V power and 500 mA current for 65 min to 120 min, depending on the size of the PCR product. To make the gel, 10× TBE buffer was used, whereas 1× TBE buffer was used to fill the tank, and the gel was stained in ethidium bromide solution. The gel was imaged using a Kodak Digital Science EDAS 290 ver. 3.6 with the Kodak ID Image analysis software ver. 3.5. Different bands of the same SSR primers were grouped according to their respective sizes by comparing the bands with a 50 bp DNA size marker ladder, and genotyping was performed visually according to the format of the different software that were used.

Statistical analyses and association mapping

The overall data were divided into two clusters to calculate the association of panicle traits with other yield traits, which was based on three different origins (East Asian, other regions of Asia, Africa), and the association of panicle traits with other yield traits, which was based on panicle pattern. All trait data and yield data were used to clarify the correlation among traits. For the statistical analysis [analyses of variances (ANOVA), *t* test, F-test, correlation coefficient and

principle component analysis (PCA)] were performed using the JMP statistical software ver. 9.0 (SAS Institute, USA). Correlation analysis was performed to elucidate the relationships among the investigated. We used the frequency distribution and a PCA to examine each trait across three different regions and 6 different PanT. Based on the field observation, the phenotypic performance was varied among the three groups (East Asian, other regions of Asian and African accession). Therefore, the *t* test was used to compare mean values, and the F-test was used to compare the variances between the three groups. ANOVA were conducted for the all traits measured using two replications in each year and over across multi years. ANOVA was carried out using the Generalized Linear Mode procedures of SAS statistical software ver. 9.0 (SAS Institute, Inc, 2010) by evaluating genotype effects for each year and across multi years. Means separation was carried out by 2 years for panicle traits and yield component traits using Fischer's protected LSD. Heritability value was calculated as a ratio of genotypic variance to the phenotypic variance. Moreover, the expected selection response was calculated using the heritability value. To understand the patterns of correlation among the inflorescence architecture and their direct and indirect effects on yield, these patterns were evaluated using a path analysis with structural equation modelling methodology (SEM). Path analyses were calculated by using the WarpPLS software ver. 3.0 (Kock 2012). The population structure among the 107 accessions was estimated using the genotype data of 98 SSR markers and the statistical package STRUCTURE ver. 2.2 (Pritchard et al. 2000) and SPAGeDi 1.3 (Hardy and Vekemans 2002). The statistical software TASSEL (Trait Analysis by Association, Evolution and Linkage) ver. 2.0.1 (Bradbury et al. 2007), with a general linear model (GLM) and a mixed linear model (MLM), were used to obtained the *P*-values, which represented the significance of LD and the fraction of total variation R² value revealed by marker effect.

Marker localisation and homology to known genes

The significant loci that correlated with the panicle architecture were physically localised by BLAST in <http://www.phytozome.net/sorghum>, <http://www.plantgdb.org/SbGDB/>, or <http://www.gramene.org/>. Markers that were previously identified as linked to

known genes were localised to the genome-based sequence information, which was provided in the Map Viewer at the NCBI website (<http://www.ncbi.nlm.nih.gov/mapview/>) and in the sorghum genome database at <http://www.phytozome.net/sorghum>. Protein sequences of genes were also used to search by BlastP, and the homologous sorghum genes were identified in <http://www.plantgdb.org/SbGDB/>.

Results

Phenotypic variations of panicle traits and other yield traits

We first examined the variability of the 11 quantitative traits that were related to the inflorescence architecture and to the yield of sorghum from the three geographic regions over 2 years to identify the characteristics that were responsible for the majority of the variation in the inflorescence architecture. The analysis of variance indicated that most of panicle traits were significantly differed at the 1 % level among accessions in two replications (Table 1). The significant mean square values obtained from ANOVA for the each trait in each year suggests that differences existed between the sorghum accessions for most panicle traits, indicating that they are highly variable (Table 1). Among the traits some characters such as PanD trait in 2010 and Pend, PanN traits in 2011 did not show strong significant level. The significant effects of accession (genotype) $\times$ (year) interaction over 2 years mean squares were shown high significant level in several components of inflorescence architecture. Moreover, we found that the patterns of observed panicle traits reflected the distribution of different origins (ESM Table 7). The mean and range of the 11 traits were shown to be widely distributed among three different geographic groups and the ranges were broad in African accessions such as Rac, PanD, MxLBZ, GNP and GWP traits. In other regions of Asian accessions, most of the traits were also widely distributed such as TotBr, PanW, and PanN. In East Asian accessions Rac and PanN traits were widely distributed. Pend, Rac, PanW, TotN and PanN traits significantly differed between East Asian and African accessions. TotN and PanN traits varied significantly between East Asian, other regions of Asian and African accessions. The African accessions had

reproducibly shorter Pend, more nodes (TotN) and longer Rac with high grain number (GNP).

Estimates of heritability and selection response

The heritability estimates is an important source of information to determine selection strategy. Heritability was calculated as the ratio of genetic variance (σ_g^2) to phenotypic variance (σ_p^2) according to (Hanson et al. 1956). We used the following formula to evaluate heritability estimates (h^2):

$$h^2 = \sigma_g^2 / \sigma_p^2 \times 100, h^2 = \frac{\sigma_g^2}{\sigma_{g/rl}^2 + \sigma_{ge/l}^2 + \sigma_g^2}, \text{ and}$$

$$\sigma_p^2 = \sigma_{e/rl}^2 + \sigma_{ge/l}^2 + \sigma_g^2$$

where σ_{ge}^2 is the variance associated with genotype $\times$ years (i.e., *genotype* $\times$ environment interaction variance) and σ_e^2 the experimental error (i.e., error variance). The terms *r* and *l* indicate the number of replication and the number of years, respectively. The heritability percentage was categorized low, moderate and high as given by Robinson et al. (1949). Moreover, the expected selection response was calculated using the heritability value to determine how a population will respond to selection. We used the general formula for selection response as ($R = i \cdot h^2 \cdot \sigma_p$). In this formulae the term *R* indicate response to selection, *i* indicate selection intensity, σ_p is the phenotypic standard deviation of trait and h^2 is heritability value. The result from the heritability estimates for each year and combined replications across years are presented in (Table 2). Replications and years played their roles in modifying the heritability estimates for different traits. Low, medium and high estimates of broad-sense heritability were found in inflorescence architecture with yield related traits (Table 2). For both growing season the high and moderate heritability obtained for most of the characters such as TotBr, Pend, PanL, TotN, GNP, GWP, MxLBZ at high heritability and Rac, PanW and PanN at moderate heritability. In 2010 high broad-sense of heritability estimates were detected for TotBr (92 %), Pend (64.8 %), PanL (70.7 %), TotN (86.9 %), GNP (85.8 %), MxLBZ (92.7 %) and GWP (84.7 %). Moderate broad-sense of heritability estimates were obtained for Rac (49.9 %), PanW (53.9 %) and PanN (54.7 %) while low heritability estimates were recorded for PanD

Table 1 Mean square of analysis of variance for inflorescence traits in two replications and across two seasons (FY 2010 and 2011) under field conditions

Source of variance	DF	TotBr	Pend	Rac	PanL	PanW	TotN
Year-1 (1st season)							
Replications	1	69.32	73.94	165	157.82	0.61	1.27
Genotype	106	536.4***	110.64***	81.19***	141.02***	3.19***	8.21***
Error	106	19.78	23.62	28.51	24.21	0.99	0.58
L.S.D (0.05)	–	8.8	9.6365	10.6	9.8	2	1.5
Year-2 (2nd season)							
Replications	1	479.5	18.84	2.8	2.07	6.5	0.02
Genotype	106	886.57***	34.41**	67.6***	125.83***	1.85***	8.03***
Error	106	135.43	17.31	22.75	8.23	0.54	1.78
L.S.D (0.05)	–	23.1	8.4	9.5	5.7	1.5	2.6
Across 2 years							
Years	1	3,879.17***	1,129.38***	51.25 ^{ns}	31.94 ^{ns}	7.29*	203.04***
Replications	2	267.97	46.39	54.1	79.94	4.52	0.65
Genotype	106	891.22***	98.38***	109.93***	183.57***	3.06***	12.58***
Years × genotype	106	476.29***	46.67***	36.35*	83.28***	1.88***	3.66***
Error	212	77.02	20.47	24.5	16.22	0.74	1.18
L.S.D (0.05)	–	12.232	6.3062	6.8996	5.6137	1.2023	1.5148
Source of variance	DF	Mean square					
		PanD	PanN	GNP	MxLBZ	GWP	
Year-1 (1st season)							
Replications	1	14.48	206.07	746,705.05	97.17	246.47	
Genotype	106	13.69*	6.3***	1,864,221.64***	120.65***	654.54***	
Error	106	9.1	1.85	142,510.44	4.57	54.34	
L.S.D (0.05)	–	5.96	2.7	748.4	4.2	14.6	
Year-2 (2nd season)							
Replications	1	0.2	0.67	846,617.23	26.12	129.49	
Genotype	106	3.1***	2.09**	514,870.3***	142.5***	114.67***	
Error	106	1.66	1.1	114,417.16	25.39	33.13	
L.S.D (0.05)	–	2.6	2.1	670.6	10	11.4	
Across 2 years							
Years	1	5.78 ^{ns}	227.44***	6,467,062.1***	551.13***	9,596.05***	
Replications	2	8.18	103.37	796,660.7	61.64	187.99	
Genotype	106	10.04***	4.14***	1,198,475.1***	208.25***	410.82***	
Years × genotype	106	8.53*	4.25***	1,180,616.7***	54.9***	358.38***	
Error	212	5.35	1.47	128,463.8	14.98	43.73	
L.S.D (0.05)	–	3.2246	1.6916	499.59	5.3953	9.2177	

n.s non-significant. Legend for quantitative traits: *TotBr* total branch per panicle, *Pend* peduncle length, *Rac* rachis length, *PanL* panicle length, *PanW* panicle width, *TotN* total node along rachis, *PanD* panicle diameter, *PanN* panicle number per plant, *GNP* grain number per panicle, *MaxLBZ* maximum length of primary branch zone, *GWP* total grain weight per panicle. Qualitative traits: *PanS* panicle shape, *PanT* panicle type, *PanEx* panicle exertion were also recorded

*, ** and *** represent significant level at $P < 0.05$, very significant at $P < 0.01$ and very highly significant at $P < 0.0001$, respectively

Table 2 Estimates of broad-sense heritability value and selection response rate for inflorescence traits in 107 SDRS in each year and across 2 years

Trait	Year	Variances					
		Genotype σ_g^2	Genotype $\times$ Years σ_{gy}^2	Error σ_e^2	Phenotype σ_p^2	Heritability (%)	Selection response R
Tot Br	2010	258.3	–	19.78	278.08	93.89	21.685
	2011	348.43	–	134.25	482.68	72.19	22.203
	Across 2 years	103.73	199.63	77.01	380.37	27.27	7.446
Pend	2010	43.507	–	23.62	67.127	64.81	7.434
	2011	8.54	–	17.31	25.85	33.04	2.351
	Across 2 years	12.92	13.09	20.46	46.47	27.80	2.653
Rac	2010	26.19	–	26.24	52.43	49.95	5.064
	2011	22.43	–	22.75	45.18	49.65	4.672
	Across 2 years	18.39	5.92	24.5	48.81	37.68	3.685
PanL	2010	58.41	–	24.21	82.62	70.70	8.996
	2011	58.79	–	8.22	67.01	87.73	10.055
	Across 2 years	25.07	33.53	16.22	74.82	33.51	4.058
PanW	2010	1.1	–	0.94	2.04	53.92	1.078
	2011	0.62	–	0.54	1.16	53.45	0.806
	Across 2 years	0.29	0.56	0.74	1.59	18.24	0.322
TotN	2010	3.81	–	0.57	4.38	86.99	2.549
	2011	3.12	–	1.78	4.9	63.67	1.973
	Across 2 years	2.23	1.23	1.18	4.64	48.06	1.4491
PanD	2010	2.26	–	9.04	11.3	20.00	0.941
	2011	1.67	–	1.66	3.33	50.15	1.281
	Across 2 years	0.37	1.58	5.32	7.27	5.09	0.192
PanN	2010	2.22	–	1.84	4.06	54.68	1.543
	2011	0.49	–	1.09	1.58	31.01	0.546
	Across 2 years	–0.025	1.38	1.47	2.825	–0.88	–0.021
GNP	2010	860,855.6	–	142,510.4	1,003,366	85.80	1,203.175
	2011	200,226.5	–	114,417.1	314,643.6	63.64	499.735
	Across 2 years	4,464.6	526,076.5	128,463.8	659,004.9	0.68	7.699
MxLBZ	2010	58.03	–	4.57	62.6	92.70	10.268
	2011	58.55	–	25.39	83.94	69.75	8.947
	Across 2 years	38.33	19.95	14.98	73.26	52.32	6.270
GWP	2010	300.1	–	54.33	354.43	84.67	22.317
	2011	40.77	–	33.12	73.89	55.18	6.640
	Across 2 years	13.11	157.32	43.73	214.16	6.12	1.254

The heritability percentage and selection response are categorized at 0–30 %: low, 31–60 %: moderate, 61 % and above: high by Robinson et al. (1949)

Legend for quantitative traits: *TotBr* total branch per panicle, *Pend* peduncle length, *Rac* rachis length, *PanL* panicle length, *PanW* panicle width, *TotN* total node along rachis, *PanD* panicle diameter, *PanN* panicle number per plant, *GNP* grain number per panicle, *MaxLBZ* maximum length of primary branch zone, *GWP* total grain weight per panicle. Qualitative traits: *PanS* panicle shape, *PanT* panicle type, *PanEx* panicle exertion were also recorded

(20 %). In 2011 high broad-sense of heritability estimates were detected for TotBr (72.2 %), PanL (87.7 %), TotN (63.7 %), GNP (64.4 %) and MxLBZ (69.7 %). Moderate broad-sense of heritability estimates were obtained for Pend (33 %), Rac (46.6 %), PanW (53.4 %), PanD (50.1 %) and PanN (31 %). However across 2 years, moderate broad-sense of heritability estimates were detected for four traits such as Rac (37.7 %), PanL (33.5 %), TotN (48.1 %), and MxLBZ (52.3 %) while low heritability estimates were recorded for TotBr (27.3 %), Pend (27.8 %), PanW (18.24 %), PanD (5 %), PanN (−88 %), GNP (0.68 %) and GWP (6.12 %).

Distribution of the different sorghum panicle types of 107 accessions based on their geographic origins

The inflorescence traits revealed six PanT classes, and these classes varied from open/loose panicles to compact elliptic panicles, indicating a high diversity of sorghum accessions among three different origins (Fig. 2). Many types of panicle existed in both diversity areas, Africa and Asia. The highest percentage of the accessions had compact type (Africa-52.94 %, East Asia-20.59 % and other regions of Asia-26.47 %), semi-compact type (Africa-23.08 %, East Asia-23.08 % and other regions of Asia-53.85 %), open type (Africa-76.47 %, East Asia-11.76 % and other regions of Asia-11.76 %), whereas a lower percentage of the accessions had broom type (Africa-25 %, East Asia-43.75 % and other regions of Asia-31.25 %) and intermediated type (Africa-54.17 %, East Asia-25 % and other regions of Asia-20.83 %) when compared between the centre of diversity areas and edge of diversity areas (Fig. 2, ESM Figure 1). Many types of panicles exist in both diversity areas, Africa and Asia. The highest percentage of the accessions had compact type (Africa-52.17 %), open type (Africa-76.47 %), broom type (East Asia-43.75 %) and with semi-compact type (other regions of Asia-53.85 %) when compared between the centre of diversity areas and the edge of diversity areas (Fig. 2).

Principal component analysis of 107 accessions based on 6 different panicle types

We considered whether the data were separated based on any of the PanT (ESM Figure 1 and Table 4). The

highest contribution and positive value of PC 1 with traits are Pend, PanW, GNP and GWP, which were substantial (i.e., $-30 > \text{loadings} > 0.30$) in 2010, and Pend, TotBr, PanD, PanW, PanEx and GWP traits, which were substantial in 2011. The loadings of PanL, TotN, MaxLBZ, PanS and PanT in 2011 were highest in PC2 in 2010, whereas PanL, TotN, MaxLBZ, PanD loadings were highest in PC2 in 2011. For PC3, the loadings of Pand, Rac, PanS, PanT, PanD, PanEx and PanN were substantial. The loadings of TotN, MaxLBZ, PanS, and PanT in 2010 and Rac, TotN, MaxLBZ and GNP in 2011 were highest in PC4. We found that PC1 appeared to separate the open type and broom type accessions from the others; however, the remaining types were not separated by PC2, PC3 and PC4. An eigenvector analysis (Table 4) suggested that the PanL, Rac, PanD, PanW, as well as the MaXLBZ, GNP and GWP, are largely responsible for this separation and are key characteristics of inflorescence architecture. Together, these results suggest that the panicle measurements that were used in this study could at least partially capture the diversity of the sorghum panicle architecture.

Phenotypic correlation among traits

The correlation coefficients (r) were calculated for all traits. The correlation of 11 quantitative traits with each other was shown in (Table 3). Focusing on the trait pairs, the correlation between two different growing seasons that are presented in Table S4 is significant ($P < 0.001$) (Table 3). In terms of panicle architecture, most of the length-based measurements showed a trend that was correlated. In particular, the PanL was positively correlated with most of the length-based measurements (Rac, MaxLBZ, PanD) and yield components (GNP and GWP). Additionally, the TotBr and TotN were also correlated with PanL. Importantly, grain yield-related traits were also positively correlated with most of the length-based measurements. In particular, the GWP was correlated with the PanW in 2010 (PanW; 0.49) and in 2011 (PanW; 0.53), with PanD in 2011 (PanD; 0.44) and with the non-length measurement, TotBr, in 2010 (TotBr; 0.38) and in 2011 (TotBr; 0.46). In addition, the GNP was correlated with the TotBr (0.39), PanW (0.43) and PanL (0.32). As expected, the GNP was highly correlated with the GWP (0.79), (0.67) in both seasons. The Pend was negatively correlated with yield-related traits (GNP; -0.32 and GWP; -0.31).

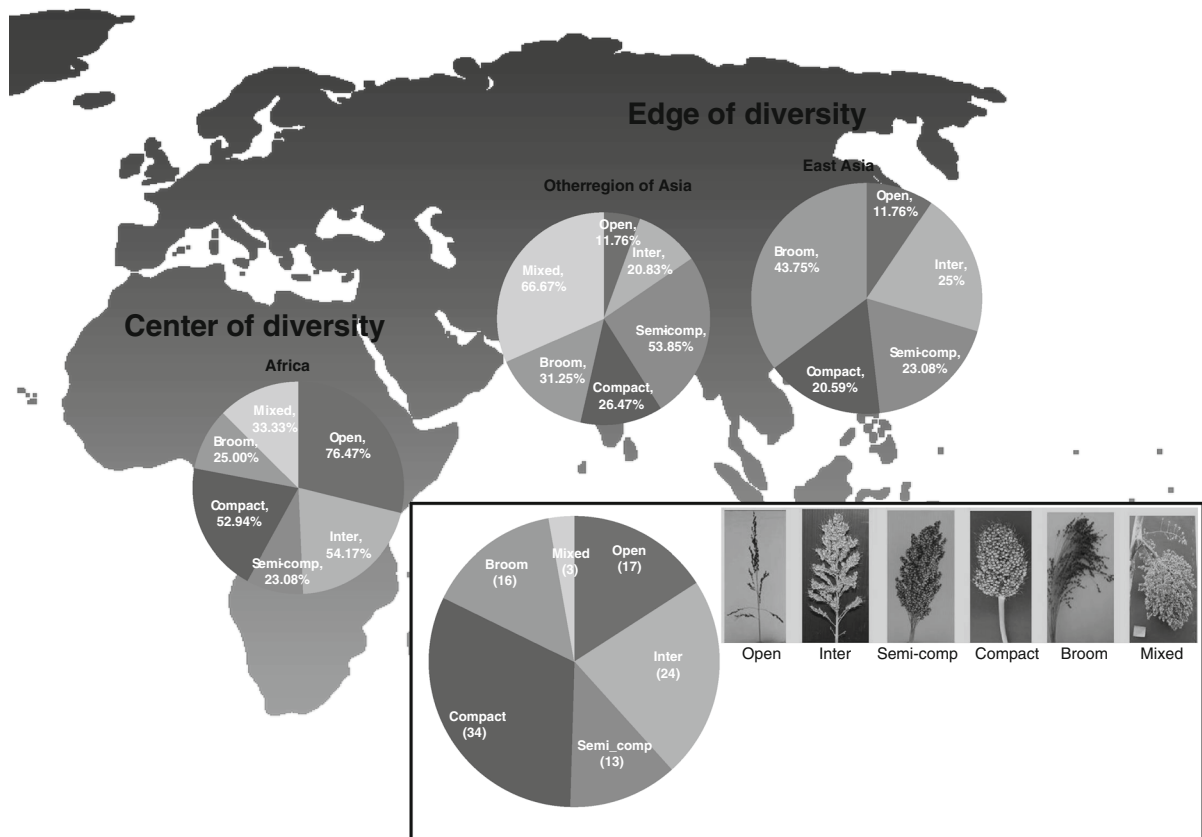


Fig. 2 Distribution of different sorghum panicle types of the 107 sorghum core collection, which is based on their geographic origins. The *large pie chart* summarises the distributions of accessions in the 107 sorghum core collection, and the *small pie charts* on the world map correspond to the country-specific distributions of sorghum accessions. The *colours* within each

small pie chart are reflective of the percentage of accessions in each origin. Legend for pie chart: *Open* open panicle type, *Inter* intermediate panicle type, *Semi-comp* semi-compact panicle type, *Compact* compact panicle type, *Broom* broom panicle type, *Mixed* mixed panicle type.

Path analysis

The path coefficient analysis indicated the specification of correlations between panicle traits and yield-related traits. This technique allowed greater quality and a complete understanding of the interactions among all measured traits. For inflorescence architecture, the direct and indirect effects on the dependent variable yield trait were analysed. The path analysis (ESM Table 4) showed a positive direct influence of the PanL on GWP because the yield was 0.31. The positive direct influences of TotBr on grain yield (0.38) and of PanD on grain yield (0.26) were also high. Grain weight (GWP) had a maximum positive direct influence on grain yield (0.54).

Population structure (Q) and inflorescence architecture

The population structure was calculated using Bayesian clustering analyses with admixture models, in which the number of populations (J) ranged from 2 to 9. Markov chain Monte Carlo (MCMC) sampling was exploited the repetition 1×10^5 times after 1×10^4 cycles of a burn-in period. The optimal number of populations was determined by the estimated logarithmic posterior probability of the Bayesian clustering. The analysis was repeated three times for each number of J . The posterior probability of $J = 3$ was the largest among other values of J (ESM Table 5). Thus, we chose $J = 3$ and obtained estimates for the

Table 3 Correlation coefficient among quantitative traits of 107 sorghum accessions

Traits	TotBr	Pend	Rac	PanL	PanW	TotN	PanD	PanN	GNP	MXL	GWP
Tot Br	1	n.s	n.s	−0.25**	0.46***	0.19*	n.s	n.s	n.s	n.s	n.s
Pend	−0.22**	1	n.s	n.s	0.19*	n.s	0.26**	n.s	n.s	n.s	−0.25**
Rac	n.s	n.s	1	0.34***	n.s	0.34***	n.s	n.s	n.s	n.s	n.s
PanL	0.29**	n.s	0.24**	1	n.s	n.s	0.34***	n.s	n.s	0.53***	n.s
PanW	0.34***	−0.20*	n.s	0.21*	1	n.s	n.s	0.34***	n.s	n.s	0.53***
TotN	n.s	n.s	0.32***	−0.35***	n.s	1	n.s	n.s	n.s	−0.44***	n.s
PanD	n.s	n.s	n.s	0.26**	0.57***	n.s	1	0.23**	0.38***	0.23**	0.44***
PanN	n.s	n.s	n.s	n.s	n.s	n.s	n.s	1	n.s	n.s	n.s
GNP	0.39***	−0.32***	n.s	0.32***	0.43***	n.s	0.24**	n.s	1	n.s	0.67***
MXL	n.s	n.s	n.s	0.52***	0.21*	−0.43***	0.21*	n.s	0.22*	1	n.s
GWP	0.38***	−0.31**	n.s	0.31**	0.49***	n.s	0.26**	n.s	0.79***	n.s	1

n.s non-significant. Legend for quantitative traits: *TotBr* total branch per panicle, *Pend* peduncle length, *Rac* rachis length, *PanL* panicle length, *TotN* total node along rachis, *PanD* panicle diameter, *PanW* panicle width, *PanN* panicle number per plant, *GNP* grain number per panicle, *MaxLBZ/MXL* maximum length of primary branch zone, *GWP* total grain weight per panicle

*, ** and *** represent significant level at $P < 0.05$, very significant at $P < 0.01$ and very highly significant at $P < 0.0001$, respectively. (*Upper diagonal* the correlation coefficient among traits for the 2011 growing season, *Lower diagonal* the correlation coefficient among traits for the 2010 growing season)

proportion of accession i 's genome that originated from population j , q_{ij} . Three sub-groups (sub-populations) were detected across the SDRS of 107 landraces (Group I, $J = 1$; Group II, $J = 2$; and Group 3, $J = 3$), which contained 35, 35 and 37 accessions, respectively, and these groups were composed of six different PanTs (ESM figure 2 (ii) and ESM Table 5).

Comparison among different models

We used the statistical software TASSEL (Trait Analysis by Association, Evolution and Linkage) ver. 2.0.1 (Bradbury et al. 2007) to obtained P values that represented the significance of LD. We compared different association models (two models from a GLM, and two models from a MLM) to evaluate the possibility of false-positives among these models by comparing observed P -values against expected P -values as described by Stich et al. (2008) and Shehzad et al. (2009b) (ESM Figure 3). Concerning GLM, we approached two different models (1) the naive model, which has no control of the population structure and kinship, and (2) the Q model, which is based on the population structure (Yu et al. 2006; Zhao et al. 2007). In MLM, we approached two different models, (1) a model that was based on kinship (K) and (2) a model that unified both the

population structure and kinship (Q + K). Among four different models, the naive model showed the highest deviation from the $y = x$ line than the other models such as K, Q and Q + K (ESM Figure 3). The K model was better than the naive model when compared; however, the results that were obtained from naive and K models detected the largest number of markers that were associated with different panicle traits among all single QTL models (data not shown). The method of both naive and K models might detect a larger number of false-positives than other models. The Q and Q + K models showed comparable results because they gave the lowest deviation from the $y = x$ line indicating these two methods might have the smallest possibility of false positive among all models. After taking consensus among models, a total of 16 loci were identified by any two models that have strong association with 14 panicle traits. After comparisons across different models K and Q models are not so bad but we selected Q + K model as the best fit model to determine sorghum panicle QTL. The selected model, Q + K, was used to test marker trait associations between 98 SSRs and 14 panicle traits because model comparisons showed that the Q + K model suitably controlled the false positive rate and gave appropriate associations with 14 panicle traits as the best model.

QTL identification

A total of 14 panicle traits were evaluated on the (SDRS) of 107 landraces population. The QTL results were presented based on the average trait values from two replications of panicle traits over two different growing seasons, 2010 and 2011 year (107 genotypes $\times$ 2 replications $\times$ 2 growing seasons). In the first growing season (Year 2010) of the data analysis revealed that 52 QTLs were detected with 14 panicle traits and the second growing season (Year 2011) of the data analysis revealed that 28 QTLs were detected at $-\text{Log}_{10} P$ value ≥ 1.5 as the threshold value, and this threshold level corresponds to the $P < 0.05$ level of significance. In case QTLs were not identified from each year with different QTLs data but those QTLs were also presented using the data to identify common QTLs. Both years of the same QTLs were presented and their chromosomal location is depicted in (Figure 3 and ESM Table 6). Sixteen loci on eight chromosomes were found to be significantly related to the patterns of observed panicle traits. Among these loci, four QTLs were responsible for length-based traits, and three QTLs were responsible for dimensional traits. Four QTLs were responsible for panicle feature such as PanT and shape, one QTL responsible for yield related trait such as grain weight. Two QTLs were responsible for branch-based traits, such as the total node number and branch number with $-\text{Log}_{10} (P)$ values ranging from 1.5 to 7.6 as the threshold value.

Marker-trait association analysis

The selected model, $Q + K$, was used to test marker trait associations between 98 SSRs and all panicle traits because model comparisons showed that the $Q + K$ model gave strong associations with ten panicle traits as the best model. The data analysis revealed that total 16 QTLs were detected with 10 panicle traits with $-\text{Log}_{10} P$ -value ranging from 1.5 to 7.6 and explained R^2 (%) ranging 0.45–41.2 % of phenotypic variation (Figs. 3, 4 and ESM Table 6). Several genomic regions affected multiple traits, including one region that affected PanL and MxLBZ (MaxLBZ). QTLs for different traits tended to be found in the same region on chromosome 4 and on chromosome 9. Additionally, QTL was involved on Chr-2, Chr-5 and Chr-6 as a novel QTL that controlled

Rac, total node number, PanD and PanT. Another single locus, *Xtxp10* (Chr-9), was found to be strongly and highly associated with two of the main panicle traits as $-\text{Log}_{10} P$ value ranging 5.6 from 7.6 and explained 0.69–22.8 % of phenotypic variation and, which included the PanL and MaxLBZ. Similarly, the single locus *Xtxp12* was highly associated with the PanL trait and was association with MaxLBZ on Chr-4. The PanL was controlled by the maximum number of SSRs on four different chromosomes (Chr-1, Chr-2, Chr-4 and Chr-9). Our results suggested that the sorghum linkage group is heavily populated with loci that were responsible for the inheritance of panicle traits of 107 genotypes, particularly the panicle elongation and branching traits. Similar results have been reported in other studies (Klein et al. 2001; Shehzad et al. 2009b; Srinivas et al. 2009). Among the QTLs, locus *Xtxp12* and *Xtxp 212* on Chr-4 was highly associated with the panicle traits and yield related trait. One of the panicle traits, the PanD, had a strong association with the single SSR locus *Xtxp8* on chromosome 2. One of the main panicle traits, the maximum length of primary branch (MaxLBZ), was controlled by a high number of SSRs on six different chromosomes. In the first growing season (Year 2010) revealed that QTLs, locus *Xtxp212* on Chr-4 was highly associated with the panicle traits and yield related trait. One of the panicle traits, the PanD, had a strong association with the single SSR locus *Xtxp25* on chromosome 2. One of the main panicle traits, the MaxLBZ, was controlled by a high number of SSRs on six different chromosomes.

Physical co-localisation of QTLs

To validate our results, we physically localised our markers and compared their positions with known genes. One QTL (*SbAGB03*), which was identified in this study, was physically localised on Chr-2 at 58,128,106 bp and was homologous to a protein-coding gene, SB02g024110, with a molecular function of binding DNA or protein. Similarly, *Xtxp8* (LG B), which was located on Chr-2 at 64,824,875 bp, was found to be within the sequence of the gene Sb02g029730, which plays an important role in ATP binding and protein tyrosine kinase activity. Further experimentation is required to establish whether either of these loci is related to drought tolerance in sorghum.

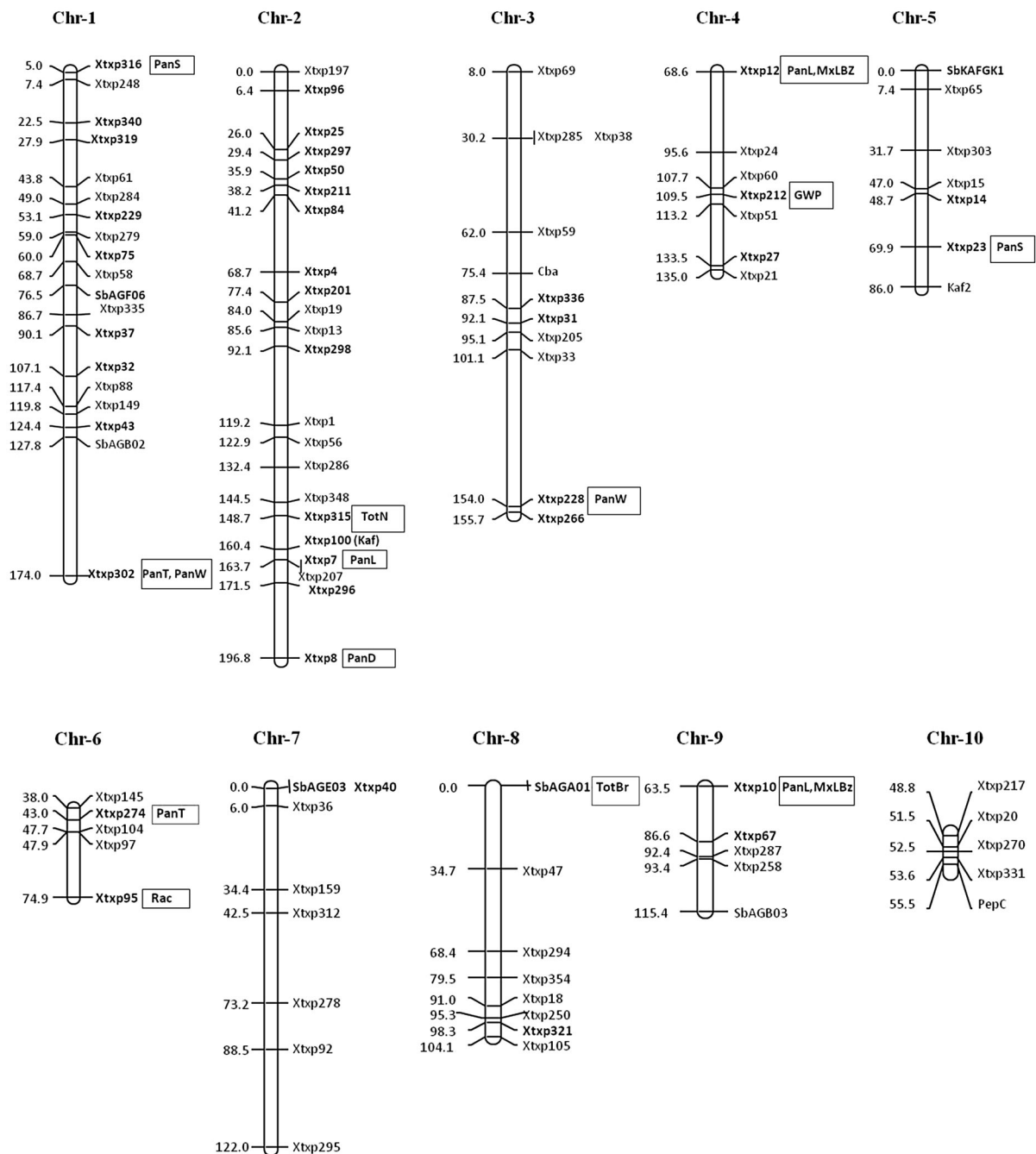


Fig. 3 QTLs for sorghum inflorescence traits and location of SSRs on ten sorghum chromosomes are shown in order as described in (Kim et al. 2005; Bhattaramakki et al. 2000; Kong et al. 2000; Taramino et al. 1997) and markers with bold face show significant associations with traits, as resolved by the Q + K model. *Legends for QTLs:* 15 QTLs identified for ten

panicle traits and yield related traits in both years are enclosed within parentheses. Legend for traits: *TotBr* total branch per panicle, *Rac* rachis length, *PanL* panicle length, *TotN* total node along rachis, *PanD* panicle diameter, *PanW* panicle width, *MxLBZ* maximum length of primary branch, *GWP* total grain weight per panicle, *PanS* panicle shape, *PanT* panicle type

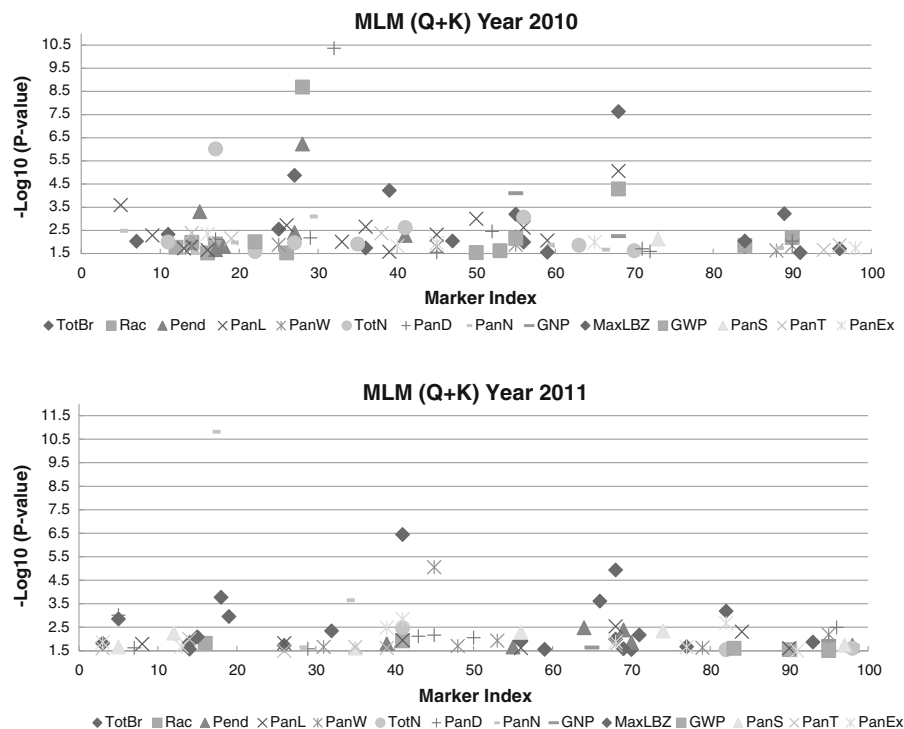


Fig. 4 Association analysis of 98 SSR markers and 14 panicle traits using the Q + K model for 107 sorghum accessions. The mixed linear model (Q + K), $-\log_{10}(P)$ values = 1.5 as threshold value (threshold level corresponds to a test at the 0.05 level of significance, respectively). Legend for traits: *Pend* peduncle length, *Rac* rachis length, *PanL* panicle length, *TotN*

total node along rachis, *TotBr* total branch per panicle, *MaxLBZ* maximum length of primary branch zone, *PanD* panicle diameter, *PanW* panicle width, *PanS* panicle shape, *PanT* panicle type, *PanEx* panicle exertion, *PanN* panicle number per plant, *GNP* grain number per panicle, and *GWP* total grain weight per panicle

Discussion

Variation in sorghum inflorescence architecture is not only the result of differences in panicle elongation but also of difference in branching and PanDs (Brown et al. 2006; Witt Hmon et al. 2013). These characteristics are helpful for breeding and botanical purposes because these panicle characters can help to separate varieties and races within a sorghum species but remain of great interest to breeders (Doust and Kellogg 2002; Doust et al. 2005; Bala et al. 1996; Futsuhara et al. 1979a, b; Kellogg 2000; Zhu et al. 2010). Sorghum classification was primarily based on the panicle pattern, grain colour and spikelet characteristics (Harlan and De Wet 1972; Snowden 1936). Cultivated sorghum is classified into five main races (bicolor, guinea, caudatum, durra and kafir). Our results explain for the link between inflorescence architecture with geographic distribution that most of the variability among main panicle traits has different

clusters between African accessions with other regions of Asian accessions and East Asian and African accessions. ANOVA could identify significant differences between East Asian and other Asian accessions for half of these traits indicated with the African accessions tending to have shorter *Pend*, more nodes (*TotN*) and longer *Rac*. These results are in agreement with a previous study (Shehzad et al. 2009a), who reported that *S. bicolor* does not have a characteristic geographical distribution between any region of Asia and Africa because it was introduced from Africa to East Asia through South Asia. Our result is in agreement with the previous report by Brown et al. (2011) that, among the five main races, bicolor can grow everywhere in Africa and Asia without any separation of geographic origin or adaptation level. For the association between the inflorescence architecture and the origin, most of the African accessions vary in that this region has more or less compact PanTs and open PanTs when compared with other regions of

Table 4 Proportions (%) of variance components for the principle components of 107 sorghum accessions across three regions

Principal component	1st	2nd	3rd	4th	Principal component	1st	2nd	3rd	4th
<i>Year 2010</i>					<i>Year 2011</i>				
Eigenvalue	3.52	2.13	1.53	1.35	Eigenvalue	2.79	2.24	1.85	1.63
Percentage contribution	25.14	15.19	10.96	9.64	Percentage contribution	19.93	15.98	13.21	11.63
Cumulative %	25.14	40.33	51.29	60.92	Cumulative %	19.93	35.91	49.12	60.75
ChiSquare	542.22	383.19	301.42	251.19	ChiSquare	507.08	413.06	334.43	264.92
<i>Traits loadings</i>					<i>Traits loadings</i>				
Pend	-0.30^a	0.29	0.37^a	-0.13	Pend	0.35^a	-0.09	-0.08	0.03
Rac	0.08	-0.29	0.29	0.10	Rac	-0.06	0.08	-0.33^a	0.50^a
PanL	0.27	0.33^a	0.09	-0.15	PanL	-0.07	0.51^a	-0.24	0.06
TotN	-0.14	-0.36^a	0.24	0.43^a	TotN	0.08	-0.30^a	0.00	0.49^a
TotBr	0.29	-0.01	0.17	0.09	TotBr	0.31^a	-0.25	0.06	-0.17
MxLBZ	0.23	0.32^a	-0.07	-0.47^a	MxLBZ	-0.07	0.48^a	-0.01	-0.34^a
PanD	0.24	0.20	0.34^a	0.13	PanD	0.42^a	0.33^a	-0.12	-0.08
PanW	0.34^a	0.14	0.26	0.24	PanW	0.41^a	0.06	0.09	-0.25
PanS	-0.16	0.45^a	-0.09	0.47^a	PanS	-0.01	0.17	0.59^a	0.22
PanT	-0.20	0.42^a	-0.18	0.40^a	PanT	0.00	0.11	0.61^a	0.11
PanEx	-0.27	0.21	0.49^a	-0.12	PanEx	0.43^a	-0.26	-0.03	-0.15
PanN	-0.10	-0.06	0.46^a	-0.13	PanN	0.10	0.03	-0.28	0.12
GNP	0.42^a	0.02	0.01	0.12	GNP	0.29	0.25	0.05	0.33^a
GWP	0.42^a	0.02	0.04	0.19	GWP	0.37^a	0.23	0.05	0.29

Legend for traits: *TotBr* total branch per panicle, *Pend* peduncle length, *Rac* rachis length, *PanL* panicle length, *TotN* total node along rachis, *PanD* panicle diameter, *PanW* panicle width, *PanN* panicle number per plant, *GNP* grain number per panicle, *MaxLBZ* maximum length of primary branch zone, *GWP* total grain weight per panicle, *PanS* panicle shape, *PanT* panicle type, *PanEx* panicle exertion

^a PC loadings larger than 0.30 and smaller than -0.30 were regarded as substantial (shown in bold)

Asian accessions and East Asian accessions (Fig. 2, ESM Figure 1). Examining the separation of varieties, which was based on PanT, the first component of the PCA was able to partially separate the broom panicle and open PanTs from the others (Table 4). This result suggests that the measurements that were used can, at least partially, capture the diversity of panicle architectures. The traits that are responsible for this separation were mainly the length-based traits: the Rac, PanL, PanD, PanW and MaXLBZ/MxLBZ, which suggests that these traits are key determinants for describing panicle diversity. Additional traits, such as Pend and TotN, are also important in further describing the structure of the panicle, and advanced methods that automatically capture the architecture using image analysis will be useful in the future.

The significant mean square values obtained across 2 years for some traits indicated that the conditions in the two different growing seasons were not similar in

some consequences of bad weather conditions and that is why the genotypes did not perform similarly in both growing seasons. The seasonal changes were shown to be the major factors affecting some traits because among traits non-significant mean square values were observed for Rac, PanL and PanD traits. Selecting for these traits might be less impact on genetic improvement. However, our result showed high significant mean squares for almost all traits in the individual analysis of variance in each year as well as the combined analysis across 2 years, suggesting that, these characters can be respond to trait selection. The variance components for the two growing seasons were shown that most of the traits had higher phenotypic and genotypic variance estimates than the environmental variance estimates except Pend in 2011, Rac and PanN in 2010 and 2011. The achievement of any breeding program depends on the greater genetic variation. The greater genetic variability

would be the higher heritability that can give the better chances to select the suitable successful generations Wannows et al. (2010). Our results showed most characters have high broad sense. For each growing season the high heritability obtained for most of the characters agreed with the findings of Arunkumar et al. (2004), Negash et al. (2005), Patel et al. (1980), Bello et al. (2007). On the other hand GWP, number of GNP and yield per panicle would be not respond to selection because of their low heritability estimates in this SDRS population. However, similar results were observed by Liang and Casady (1966), Liang and Walter (1968), Singh and Singh (1973), and Bello et al. (2001). They reported that the low heritability estimate of grain yield is due to the direct or indirect multiplicative effects of several yield components on grain yield. However, our results revealed that the SDRS genotypes have a broad base genetic background as well as good potential that will respond positively to selection. Therefore, most of the panicle traits and yield related traits expressions in SDRS can be developed in breeding programs.

Examining the correlations between the measured traits, we observed that many of the length-based elongation measurements showed positive correlations. In terms of the correlation between yield and panicle architecture, we found that the PanW, PanL, TotBr and Pend correlated with the grain yield (GWP). Our results agree with previous studies by Saeed and Francis (1983) and Maman et al. (2004), who reported that the yield per plant and panicle head length were highly correlated. Interestingly, the PanD was also positively correlated with the grain number (GNP) and grain weight (GWP). Our results for grain yield are in agreement with another previous study by Maman et al. (2004). GWP and GWP were also significantly correlated with PanD and PanL, as well as high coefficients of genotypic correlation between PanL and other traits, such as the total number of branch per panicle, which have been reported many authors (Brown et al. 2006; Srinivas et al. 2009). The path analysis indicated that the PanL had a positive effect on yield-related traits. Moreover, TotBr and PanD also showed a highly effect on yield (ESM Table 4). This result revealed that the trait selection for PanL, total branch number and PanD had a strong impact on grain yield because we have shown that many of these length-based measurements are correlated, which may suggest a common genetic regulation, and that several

traits are likely to influence yield. Our result revealed that the PanL, PanD, PanW, TotN and TotBr are important component traits for the variation in sorghum inflorescence architecture and also that these traits contribute directly and indirectly to yield improvement. Plotting the STRUCTURE results on the inflorescence/PanTs revealed association of subgroup membership (broom, open and compact) probability. These sub-populations differed in three groups of inflorescence/PanTs from six different PanTs, which were associated with 14 inflorescence traits. The cluster dendrograms of 107 accessions was color-coded based on the phenotypic data results, and our results are generally in agreement with the STRUCTURE plot classification of three sub-populations (Fig. 2 and ESM Table 5). Cultivated sorghum is classified into five main races and according to their division identification system, broom corn generally falls into bicolor type, bicolor and guinea races have open panicles, kafir and durra races have compact panicles, and caudatum PanTs can vary (Harlan and De Wet 1972). In previous successful reports by Brown et al. (2011) and by Casa et al. (2008), the phenotype-based racial classification of sorghum, which was based on panicle and spikelet characteristics, is controlled by a limited number of genomic regions; however, in genotype-based classification using STRUCTURE analysis, sorghum races can be classified, except for bicolor because it is based on random markers that are distributed across the genome, which can capture the genomic variation among sorghum races. In this study, an SDRS of 107 genotypes generally formed a sub-population by STRUCTURE classification, which was based on the inflorescence architecture, but did not clearly distinguish a specific major group. Thus, our results indicate that sorghum PanTs that are distributed throughout the African and Asian regions most likely vary according to different adaptation levels, temperature, humidity and rainfall patterns. The diversity in PanS and compactness are likely to indicate the selection of varieties that can survive in different environments. However, our result is in agreement with the previous report by Brown et al. (2011), which found that, among the five main races, bicolor, can grow everywhere in Africa and Asia but does not form a clear, separate sub-population.

In total, 16 loci were detected using 98 SSRs mapped markers. The total 16 QTLs were distributed

among 8 chromosomes. Our results showed the locations of QTLs for 14 panicle traits in each year as four types: length-based traits (Rac, PanL, MaxLBZ), branching-based traits (TotN, TotBr), size and dimensional traits (PanD, PanW, PanT and PanS) and panicle-related yield traits (GWP and GNP) (Fig. 4 and ESM Table 6). Our present research examined characteristics of panicle architecture and gene influence of these features. Among these characteristics, PanL has been investigated by many previous reports; however, the PanL consists of the Rac and the top most primary branches on the rachis. In this study, we analysed PanL by separating these different parts. Interestingly, although many QTLs for PanL (ten QTLs in 2010, four QTLs in 2011, four QTLs in both years) were detected on six chromosomes, our results did not reveal whether PanL itself and Rac are controlled by different QTLs. No QTLs of the latter were detected together. Our result indicated that these traits appear to be under separate genetic control (from length based traits) in sorghum inflorescence architecture because there is no overlap between QTLs that were detected for these traits. In elongation, the characteristics of Rac and branches along the rachis, such as MaxLBZ, also appear to be distinct processes. The association between QTLs for PanL and MaxLBZ on Chr-4 and Chr-9 (LG- F) accounts for the strong correlation between these traits. In this case, the allele that was associated with greater PanL was associated with increased primary branch length. Shehzad et al. (2009b) reported that the same location of two QTLs of PanL traits (*SbAGF06* on Chr-1 and *Xtxp7* on Chr-2) matched with the same chromosome location and the same QTL of the length-based trait PanL, which was identified in the study. More than one QTL for different traits tended to be found in the same region on chromosome 4 (linkage group D). Earlier studies on inflorescence architecture in sorghum, rice and maize have suggested that branching characteristics have the most important role in panicle characteristics. QTLs for these traits were in the same position. Our results of GWAS analysis demonstrated that PanL and MaxLBZ might be under the same genetic control. The highly significant association of the PanD on Chr-2 (LG-B) has not been previously reported in linkage mapping studies. Additionally, QTLs for that trait in this location have not been detected by the previous analysis. The genomic region on Chr-4 for QTLs showed an interaction with not only the panicle traits

but also yield-related traits, such as the total grain weight. Panicle dimensional traits, such as the PanD and width (PanD, PanW), are important traits for panicle improvement (i.e., dense panicles leads to low grain quality due to a disorder in panicle dimension and size). It is worth noting that we were unable to identify novel QTLs that were suggested to be potentially able to control PanD and PanW on three chromosomes (Chr-1, Chr-2 and Chr-3). Additionally, QTL was involved on Chr-1 for PanW, Chr-2 for TotN and PanD, Chr- 6 for Rac, Chr-10 for PanT as new QTLs that controlled inflorescence architecture. In each growing season the effects and relative positions of PanL, PanD, TotBr and Rac were in accordance with the QTL distribution of several inter-related other panicle and yield component traits, such as MaxLBZ, TotN, GWP and PanN traits. Several genomic regions affected multiple traits, including one region that affected PanL, Rac, MaxLBZ and PanN and another region that influenced PanD, TotBr and GWP. Nine QTLs were found for the TotBr on six chromosomes consisting of three on Chr-1, two on Chr-2, and others on Chr-3, Chr-4, Chr-7 and Chr-8. Among these QTLs, the QTL regions on Chr-1 with locus (*Xtxp43*) and locus (*Xtxp40*) on Chr-7 matched with similar positions (by using RIL population) to a previous report by Srinivas et al. (2009). Similarly, the two QTLs were mapped for TotBr in the study at similar position as the branch length QTL, which was located on Chr-1 (100 cM) and on Chr-3 by Brown et al. (2006). These results demonstrate that the branch length QTLs are common in this study and in earlier studies on the branch trait in sorghum. The branch length QTL (MaxLBZ) on Chr-3 was matched with a similar position as the branch length QTL that was reported by Brown et al. (2006). In the first growing season data (Year 2010) revealed five positive height associations, *Xtxp25*, *Xtxp297*, *Xtxp50*, *Xtxp211* and *Xtxp84*, were on the same chromosome, approximately 10 bp apart. According to this result of the QTL analysis of panicle elongation traits showed major QTL regions that formed as a cluster that was located on Chr-2. These regions overlapped completely with clustered QTLs. The results suggest that the sorghum linkage groups are heavily populated with loci that are responsible for the inheritance of panicle elongation and dimensional traits of SDRS genotypes. GWAS analyses on inflorescence architecture can identify novel loci on these locations potentially with the key traits for sorghum

inflorescence architecture. This location and these traits have not been detected by the previous analyses. These trait characteristics can be used for selection and classification purposes by simplify procedures of evaluations. For both the PanD QTLs on Chr-2, the allele that was associated with a greater PanD is associated with increased node number along longer PanL similar to the loose PanT. Other panicle component traits, such as MaxLBZ and PanL, were found on chromosome 9 and chromosome 4 (LG-F and LG-D) similar to the most significant QTL in this study. In the Poaceae family, sorghum has been much less studied for genes affecting inflorescence architecture than other cereal crops. Brown et al. (2006) have mapped two QTLs for panicle primary branch number and a single QTL for secondary branching number while characterising the inflorescence architecture in sorghum. Srinivas et al. (2009) have mapped five QTLs that were detected for branching patterns. Other QTLs for Rac, PanD, PanW, TotN, MaxLBZ, PanS and PanT in the study did not relate to any of the QTL for panicle branching patterns that were identified in previous reports. Therefore, it is likely that these QTLs may be new loci that regulate the panicle and its component traits in sorghum, which results in their involvement in inflorescence architecture. These results will serve as preliminary findings of QTLs for the genetic basis of the inflorescence architecture and component traits, and the further evaluation of the germplasm for these traits is in progress. However, this study will provide other possibilities for more detailed studies, such as increasing the population size and saturating the target genomic regions by adding large-scale molecular markers with powerful molecular techniques (i.e., next generation sequencing, genome-wide SNP discovery, whole genome re-sequencing and map-based cloning of the gene underlying the QTLs) would be required in future studies. These findings can lead to the emergence of a new era of sorghum genomics, which are aimed at bridging the knowledge gap between genotype and phenotype in sorghum inflorescence architecture.

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